[750] (CONTINUED)

ON THE THEORY OF RECIPROCAL SURFACES.

601. In part explanation, observe that the definitions of ρ and σ agree with those already given. The nodal torse is the torse enveloped by the tangent planes along the nodal curve; if the nodal curve meets the curve of contact a, then a tangent plane of the nodal torse passes through the arbitrary point, that is, ρ will be the number of these planes which pass through the arbitrary point, viz. the class of the torse. So also the cuspidal torse is the torse enveloped by the tangent planes along the cuspidal curve; and σ will be the number of these tangent planes which pass through the arbitrary point, viz. it will be the class of the torse. Again, as regards ρ' and σ': the node-couple torse is the envelope of the bitangent planes of the surface, and the node-couple curve is the locus of the points of contact of these planes. Similarly, the spinode torse is the envelope of the parabolic planes of the surface, and the spinode curve is the locus of the points of contact of these planes, viz. it is the curve UH of intersection of the surface and its Hessian; the two curves are the reciprocals of the nodal and the cuspidal torses respectively, and the definitions of ρ', σ' correspond to those of ρ and σ.

602. In regard to the nodal curve b, we consider k the number of its apparent double points (excluding actual double points); f the number of its actual double points (each of these is a point of contact of two sheets of the surface, and there is thus at the point a single tangent plane, viz. this is a plane f', and we thus have $f' = f$); t the number of its triple points; and j the number of its pinch-points: these last are not singular points of the nodal curve *per se*, but are singular in regard to the curve as nodal curve of the surface, viz. a pinch-point is a point at which the two tangent planes are coincident. The curve is considered as not having any stationary points other than the points γ, which lie also on the cuspidal curve; and the expression for the class consequently is $q = b^2 - b - 2k - 2f - 3\gamma - 6t$.

603. In regard to the cuspidal curve c, we consider h the number of its apparent double points; and upon the curve, not singular points in regard to the curve *per se*, but only in regard to it as cuspidal curve of the surface, certain points in number β, χ, ω respectively. The curve is considered as not having any actual double or other multiple points, and as not having any stationary points except the points β, which lie also on the nodal curve; and the expression for the class consequently is

$$r = c^2 - c - 2h - 3\beta.$$

604. The γ are points where the cuspidal curve with the two sheets (or say rather half-sheets) belonging to it are intersected by another sheet of the surface; the curve of intersection with such other sheet, belonging to the nodal curve of the surface, has evidently a stationary (cuspidal) point at the point of intersection.

As to the points β, to facilitate the conception, imagine the cuspidal curve to be a semi-cubical parabola, and the nodal curve a right line (not in the plane of the curve) passing through the cusp; then intersecting the two curves by a series of parallel planes, any plane which is, say, above the cusp, meets the parabola in two real points and the line in one real point, and the section of the surface is a curve with two real cusps and a real node; as the plane approaches the cusp, these approach

together, and, when the plane passes through the cusp, unite into a singular point in the nature of a triple point (= node + two cusps); and when the plane passes below the cusp, the two cusps of the section become imaginary, and the nodal line changes from crunodal to acnodal.

605. At a point i the nodal curve crosses the cuspidal curve, being on the side away from the two half-sheets of the surface acnodal, and on the side of the two half-sheets crunodal, viz.

the two half-sheets intersect each other along this portion of the nodal curve. There is at the point a single tangent plane, which is a plane i'; and we thus have $i = i'$.

606. As already mentioned, a cnicnode C is a point where, instead of a tangent plane, we have a tangent quadri-cone; at a binode B, the quadri-cone degenerates into a pair of planes. A cnictrope C' is a plane touching the surface along a conic; in the case of a bitrope B', the conic degenerates into a flat conic or pair of points.

607. In the original formulae for $a(n-2)$, $b(n-2)$, $c(n-2)$, we have to write $\kappa - B$ instead of κ, and the formulae are further modified by reason of the singularities θ and ω. So, in the original formulae for $a(n-2)(n-3)$, $b(n-2)(n-3)$, $c(n-2)(n-3)$, we have instead of δ to write $\delta - C - 3\omega$, and to substitute new expressions for $[ab]$, $[ac]$, $[bc]$; viz. these are

$$[ab] = ab - 2\rho - j,$$

$$[ac] = ac - 3\sigma - \chi - \omega,$$

$$[bc] = bc - 3\beta - 2\gamma - i.$$

The whole series of equations thus is

(1) $a' = a.$
(2) $f' = f.$
(3) $i' = i.$
(4) $\alpha = n(n-1) - 2b - 3c.$
(5) $\kappa = 3n(n-2) - 6b - 8c.$
(6) $\delta = \frac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \frac{9}{2}c(c-1).$
(7) $a(n-2) = \kappa - B + \rho + 2\sigma + 3\omega.$
(8) $b(n-2) = \rho + 2\beta + 3\gamma + 3t.$
(9) $c(n-2) = 2\sigma + 4\beta + \gamma + \theta + \omega.$
(10) $a(n-2)(n-3) = 2(\delta - C - 3\omega) + 3(ac - 3\sigma - \chi - 3\omega) + 2(ab - 2\rho - j).$
(11) $b(n-2)(n-3) = 4k + (ab - 2\rho - j) + 3(bc - 3\beta - 2\gamma - i).$
(12) $c(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i).$
(13) $q = b^2 - b - 2k - 2f - 3\gamma - 6t.$
(14) $r = c^2 - c - 2h - 3\beta.$

Also, reciprocal to these,

(15) $a' = n'(n'-1) - 2b' - 3c'.$
(16) $\kappa' = 3n'(n'-2) - 6b' - 8c'.$
(17) $\delta' = \frac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + 2b'(b'-1) + 6b'c' + \frac{9}{2}c'(c'-1).$
(18) $a'(n'-2) = \kappa' - B' + \rho' + 2\sigma' + 3\omega'.$
(19) $b'(n'-2) = \rho' + 2\beta' + 3\gamma' + 3t'.$
(20) $c'(n'-2) = 2\sigma' + 4\beta' + \gamma' + \theta' + \omega'.$
(21) $a'(n'-2)(n'-3) = 2(\delta' - C' - 3\omega') + 3(a'c' - 3\sigma' - \chi' - 3\omega') + 2(a'b' - 2\rho' - j').$
(22) $b'(n'-2)(n'-3) = 4k' + (a'b' - 2\rho' - j') + 3(b'c' - 3\beta' - 2\gamma' - i').$
(23) $c'(n'-2)(n'-3) = 6h' + (a'c' - 3\sigma' - \chi' - 3\omega') + 2(b'c' - 3\beta' - 2\gamma' - i').$
(24) $q' = b'^2 - b' - 2k' - 2f' - 3\gamma' - 6t'.$
(25) $r' = c'^2 - c' - 2h' - 3\beta',$

together with one other independent relation: in all 26 relations between the 46 quantities.

608. The new relation may be presented under several different forms, equivalent to each other in virtue of the foregoing 25 relations; these are

$$(26)\ \ 2(n-1)(n-2)(n-3) - 12(n-3)(b+c) + 6q + 6r + 24t + 42\beta + 30\gamma + \tfrac{8}{3}\theta = \Sigma,$$

$$(27)\ \ 26n - 12c - 4C - 10B + \beta - 7j - 8\chi + \tfrac{2}{3}\theta - 4\omega = \Sigma;$$

in each of which two equations Σ is used to denote the same function of the accented letters that the left-hand side is of the unaccented letters.

$$\begin{aligned}(28)\ \ \beta' + \tfrac{2}{3}\theta' = {} & 2n(n-2)(11n-24)\\ & + (-66n+184)\,b\\ & + (-93n+252)\,c\\ & + 22(2\beta+3\gamma+3t)\\ & + 27(4\beta+\gamma+\theta)\\ & + \beta + \tfrac{2}{3}\theta\\ & - 24C - 28B - 27j - 38\chi - 73\omega\\ & + 4C' + 10B' + 7j' + 8\chi' - 4\omega'.\end{aligned}$$

Or, reciprocally,

$$\begin{aligned}(29)\ \ \beta + \tfrac{2}{3}\theta = {} & 2n'(n'-2)(11n'-24)\\ & + (-66n'+184)\,b'\\ & + (-93n'+252)\,c'\\ & + 22(2\beta'+3\gamma'+3t')\\ & + 27(4\beta'+\gamma'+\theta')\\ & + \beta' + \tfrac{2}{3}\theta'\\ & - 24C' - 28B' - 27j' - 38\chi' - 73\omega'\\ & + 4C + 10B + 7j + 8\chi - 4\omega.\end{aligned}$$

The equation (26) expresses that the surface and its reciprocal have the same deficiency; viz. the expression for the deficiency is

$$\begin{aligned}(30)\ \ \text{Deficiency} &= \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-3)(b+c) + \tfrac{1}{2}(q+r)\\ &\quad + 2t + \tfrac{7}{2}\beta + \tfrac{5}{2}\gamma + \tfrac{2}{9}\theta,\\ &= \tfrac{1}{6}(n'-1)(n'-2)(n'-3) - \&\text{c}.\end{aligned}$$

609. The equation (28) (due to Prof. Cayley) is the correct form of an expression for β', first obtained by him (with some errors in the numerical coefficients) from independent considerations. But it is best obtained by means of the equation (26); and (27) is a relation presenting itself in the investigation. In fact, considering α as standing for its value $n(n-1) - 2b - 3c$, we have from the first 25 equations

$$\begin{array}{rll} 6 & \alpha & = \Sigma,\\ +2 & 3n - c - \kappa & = \Sigma,\\ -2 & a(n-2) - \kappa + B - \rho - 2\sigma - 3\omega & = \Sigma,\\ -4 & b(n-2) - \rho - 2\beta - 3\gamma - 3t & = \Sigma,\\ -6 & c(n-2) - 2\sigma - 4\beta - \gamma - \theta - \omega & = \Sigma,\\ +2 & n + \kappa - \sigma - 2\beta - 4\gamma - 2j - 3\chi - 3\omega & = \Sigma,\\ -3 & 2q - 2\rho + \beta + j & = \Sigma,\\ -2 & 3r + c - 5\sigma - \beta - 4\gamma + \chi - \omega & = \Sigma;\end{array}$$

multiplying these equations by the numbers set opposite to them respectively, and adding, we find

$$\begin{aligned}&-2n^3+12n^2+4n+b(12n-36)+c(12n-48)\\&-6q-6r-4C-10B-41\beta-30\gamma-24t-7j-8\chi+\tfrac{2}{3}\theta-4\omega=\Sigma,\end{aligned}$$

and adding hereto (26) we have the equation (27); and from this (28), or by a like process, (29), is obtained without much difficulty. As to the 8 Σ-equations or symmetries, observe that the first, third, fourth, and fifth are in fact included among the original equations (for an expression which vanishes is in fact $=\Sigma$); we have from them moreover $3n-c=3a'-\kappa'$, and thence $3n-c-\kappa=3a'-\kappa'-\kappa$, which is $=\Sigma$, or we have thus the second equation; but the sixth, seventh, and eighth equations have yet to be obtained.

610. The equations (15), (16), (17) give

$$\begin{aligned}n'&=\alpha(\alpha-1)-2\delta-3\kappa,\\c'&=3\alpha(\alpha-2)-6\delta-8\kappa,\\b'&=\tfrac{1}{2}\alpha(\alpha-2)(\alpha^2-9)-(\alpha^2-\alpha-6)(2\delta+3\kappa)+2\delta(\delta-1)+6\delta\kappa+\tfrac{9}{2}\kappa(\kappa-1).\end{aligned}$$

From (7), (8), (9), we have

$$\begin{aligned}(\alpha-b-c)(n-2)&=\kappa-B-6\beta-4\gamma-3t-\chi+2\omega,\\(\alpha-2b-3c)(n-2)(n-3)&=2(\delta-C)-8k-18h-6bc+18\beta+12\gamma+6i-6\omega;\end{aligned}$$

substituting these values for κ and δ, and for α its value $=n(n-1)-2b-3c$, we obtain the values of n', c', b'; viz. the value of n' is

$$\begin{aligned}n'=n(n-1)^2&-n(7b+12c)+4b^2+8b+9c^2+15c\\&-8k-18h+18\beta+12\gamma+12i-9t\\&-2C-3B-3\omega.\end{aligned}$$

Observe that the effect of a cnicnode C is to reduce the class by 2, and that of a binode B to reduce it by 3.

611. We have

$$(n-2)(n-3)=n^2-n+(-4n+6)=\alpha+2b+3c+(-4n+6);$$

making this substitution in the equations (10), (11), (12), which contain $(n-2)(n-3)$, these become

$$\begin{aligned}a(-4n+6)&=2(\delta-C)-\alpha^2-4\rho-9\sigma-2j-3\chi-15\omega,\\b(-4n+6)&=4k-2b^2-9\beta-6\gamma-3t-2\rho-j,\\c(-4n+6)&=6h-3c^2-6\beta-4\gamma-2i-3\sigma-\chi-3\omega,\end{aligned}$$

which are the foregoing equations (C); adding to each equation four times the corresponding equation with the factor $(n-2)$, these become

$$\begin{aligned}\alpha^2-2\alpha&=2(\delta-C)+4(\kappa-B)-\sigma-2j-3\chi-3\omega,\\2b^2-2b&=4k-\beta+6\gamma+12t-3t+2\rho-j,\\3c^2-2c&=6h+10\beta+4\theta-2i+5\sigma-\chi+\omega.\end{aligned}$$

Writing in the first of these $\alpha^2 - 2\alpha = n + 2\delta + 3\kappa - \alpha$, and reducing the other two by means of the values of q, r, the equations become

$$n - \alpha = -2C - 4B + \kappa - \sigma - 2j - 3\chi - 3\omega,$$
$$2q + \beta + 3t + j = 2\rho,$$
$$3r + c + 2i + \chi = 5\sigma + \beta + 4\theta + \omega,$$

which give at once the last three of the 8 Σ-equations.

The reciprocal of the first of these is

$$\sigma' = \alpha - n + \kappa' - 2j' - 3\chi' - 2C' - 4B' - 3\omega',$$

viz. writing herein

$$\alpha = n(n-1) - 2b - 3c \quad \text{and} \quad \kappa' = 3n(n-2) - 6b - 8c,$$

this is

$$\sigma' = 4n(n-2) - 8b - 11c - 2j' - 3\chi' - 2C' - 4B' - 3\omega',$$

giving the order of the spinode curve; viz. for a surface of the order n without singularities, this is $= 4n(n-2)$, the product of the orders of the surface and its Hessian.

612. Instead of obtaining the second and third equations as above, we may to the value of $b(-4n+6)$ add twice the value of $b(n-2)$; and to twice the value of $c(-4n+6)$ add three times the value of $c(n-2)$, thus obtaining equations free from ρ and σ respectively; these equations are

$$b(-2n+2) = 4k - 2b^2 - 5\beta - 3t + 6t - j,$$
$$c(-5n+6) = 12h - 3c^2 - 6\gamma - 4i - 2\chi + 3\theta - 3\omega,$$

equations which, introducing therein the values of q and r, may also be written

$$b(2n-4) + 6\gamma + 6t + 3t + j + 4f = 2q + 5\beta$$
$$c(5n-12) + 3\theta = 6r + 18\beta + 5\gamma + 4i + 2\chi + 3\omega.$$

Considering as given, n the order of the surface; the nodal curve, with its singularities b, k, f, t; the cuspidal curve, with its singularities c, h; and the quantities β, γ, i which relate to the intersections of the nodal and cuspidal curves; the first of the two equations gives j, the number of pinch-points, being singularities of the nodal curve, *quoad* the surface; and the second equation establishes a relation between θ, χ, ω, the numbers of singular points of the cuspidal curve *quoad* the surface.

In the case of a nodal curve only, if this be a complete intersection $P = 0$, $Q = 0$, the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, and the first equation is

$$b(-2n+2) = 4k - 2b^2 + 6t - j,$$

or, assuming $t = 0$, say $j = 2(n-1)b - 2b^2 + 4k$, which may be verified; and so in the case of a cuspidal curve only, when this is a complete intersection $P = 0$, $Q = 0$, the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, where $AC - B^2 = MP + NQ$; and the second equation is

$$c(-5n+6) = 12h - 6c^2 - 2\chi + 3\theta - 3\omega,$$

or, say $2\chi + 3\omega = (5n-6)c - 6c^2 + 12h + 3\theta$, which may also be verified.

613. We may in the first instance out of the 46 quantities consider as given the 14 quantities

$$n \;:\; b, k, f, t \;:\; c, h, \theta, \chi \;:\; \beta, \gamma, i \;:\; C, B;$$

then of the 26 relations, 17 determine the 17 quantities

$$\alpha, B, \kappa, \rho, \delta \ : \ j, q \ : \ r, \omega$$
$$n' \ : \ a', B', \kappa' \ : \ b', f' \ : \ c' \ : \ i'$$

and there remain the 9 equations

$$(18), (19), (20), (21), (22), (23), (24), (25), (28),$$

connecting the 15 quantities

$$\rho', \sigma' \ : \ k', t', j', q' \ : \ h', \theta', \chi', \omega', r' \ : \ \beta', \gamma' \ : \ C', B'.$$

Taking then further as given the 5 quantities j', χ', t', C', B',

$$\begin{array}{rl} \text{equations (18) and (21) give} & \rho',\ \sigma', \\ \text{equation (19)} & \text{gives } 2\beta' + 3\gamma' + 3t', \\ (20) & \text{gives } 4\beta' + \gamma' + \theta', \\ (28) & \text{gives } \beta' + \tfrac{2}{3}\theta', \end{array}$$

so that, taking also t' as given, these last three equations determine β', γ', θ'; and finally

$$\begin{array}{rl} \text{equation (22) gives} & k', \\ (23) & \text{gives } h', \\ (24) & \text{gives } q', \\ (25) & \text{gives } r', \end{array}$$

viz. taking as given in all 20 quantities, the remaining 26 will be determined.

614. In the case of the general surface of the order n, without singularities, we have as follow

$$\begin{aligned}
n &= n, \\
a &= n(n-1), \\
\delta &= \tfrac{1}{2}n(n-1)(n-2)(n-3), \\
\kappa &= n(n-1)(n-2), \\
n' &= n(n-1)^2, \\
a' &= n(n-1), \\
\delta' &= \tfrac{1}{2}n(n-2)(n^2-9), \\
\kappa' &= 3n(n-2), \\
b' &= \tfrac{1}{2}n(n-1)(n-2)(n^3-n^2+n-12), \\
k' &= \tfrac{1}{8}n(n-2)(n^7-6n^6+16n^5-54n^4+164n^3-288n^2 \\
&\qquad\qquad + 547n^2 - 1058n^3 + 1068n^2 - 1214n + 1464), \\
h' &= \tfrac{1}{2}n(n-2)(n^5-4n^4+7n^3-45n^2+114n^2-111n^2+548n-960), \\
q' &= n(n-2)(n-3)(n^2+2n-4), \\
\rho' &= n(n-2)(n^3-n^2+n-12), \\
c' &= 4n(n-1)(n-2), \\
h' &= \tfrac{1}{2}n(n-2)(16n^4-64n^3+80n^2-108n+156), \\
i' &= 2n(n-2)(3n-4), \\
\sigma' &= 4n(n-2), \\
\beta' &= 2n(n-2)(11n-24), \\
\gamma' &= 4n(n-2)(n-3)(n^2-3n+16),
\end{aligned}$$

the remaining quantities vanishing.

615. The question of singularities has been considered under a more general point of view by Zeuthen, in the memoir "Recherche des singularités qui ont rapport à une droite multiple d'une surface," *Math. Annalen*, t. IV. (1871), pp. 1–20. He attributes to the surface:

A number of singular points, viz. points at any one of which the tangents form a cone of the order μ, and class ν, with $y+\eta$ double lines, of which y are tangents to branches of the nodal curve through the point, and $z+\zeta$ stationary lines, whereof z are tangents to branches of the cuspidal curve through the point, and with u double planes and v stationary planes; moreover, these points have only the properties which are the most general in the case of a surface regarded as a locus of points; and Σ denotes a sum extending to all such points. (The foregoing general definition includes the cnicnodes $\mu=\nu=2$, $y=\eta=z=\zeta=u=v=0$, and the binodes $\mu=2$, $\eta=1$, $z=y=\&\text{c.}=0$.)

And, further, a number of singular planes, viz. planes any one of which touches along a curve of the class μ' and order ν', with $y'+\eta'$ double tangents, of which y' are generating lines of the node-couple torse, $z'+\zeta'$ stationary tangents, of which z' are generating lines of the spinode torse, u' double points and v' cusps; it is, moreover, supposed that these planes have only the properties which are the most general in the case of a surface regarded as an envelope of its tangent planes; and Σ' denotes a sum extending to all such planes. (The definition includes the cnictropes $\mu'=\nu'=2$, $y'=\eta'=z'=\zeta'=u'=v'=0$, and the bitropes $\mu'=2$, $\eta'=1$, $z'=y'=\&\text{c.}=0$.)

616. This being so, and writing

$$x=\nu+2\eta+3\zeta,\quad x'=\nu'+2\eta'+3\zeta',$$

the equations (7), (8), (9), (10), (11), (12), contain, in respect of the new singularities, additional terms, viz. these are

$$\begin{aligned}
a(n-2) &= \cdots+\Sigma\,[x(\mu-2)-\nu-2\zeta],\\
b(n-2) &= \cdots+\Sigma\,[y(\mu-2)],\\
c(n-2) &= \cdots+\Sigma\,[z(\mu-2)],\\
a(n-2)(n-3) &= \cdots+\Sigma\,[x(-\nu+7)+2\eta+4\zeta],\\
b(n-2)(n-3) &= \cdots+\Sigma\,[y(-\nu+8)]-\Sigma'\,(4u'+3v'),\\
c(n-2)(n-3) &= \cdots+\Sigma\,[z(-\nu+9)]-\Sigma'\,(2u'),
\end{aligned}$$

and there are of course the reciprocal terms in the reciprocal equations (18), (19), (20), (21), (22), (23). These formulae are given without demonstration in the memoir just referred to: the principal object of the memoir, as shown by its title, is the consideration not of such singular points and planes, but of the multiple right lines of a surface; and in regard to these, the memoir should be consulted.

751.

NOTE ON RIEMANN'S PAPER "VERSUCH EINER ALLGEMEINEN AUFFASSUNG DER INTEGRATION UND DIFFERENTIATION*."

[From the *Mathematische Annalen*, t. XVI. (1880), pp. 81, 82.]

The Editors of Riemann's works remark that the paper in question was contained in a MS. of his student time (dated 14 Jan. 1847) and was probably never intended for publication: indeed that he would not in later years have recognised the validity of the principles upon which it is founded. The idea is however a noticeable one: Riemann considers Z_{x+h} a function of $x+h$, expanded in a doubly infinite, necessarily divergent, series of integer or fractional powers of h, according to the law

$$Z_{x+h} = \sum_{\nu=-\infty}^{\nu=+\infty} k_\nu\, x \cdot h^\nu, \tag{2}$$

where the meaning is explained to be that the exponents differ from each other by integer values, in effect, that ν has all the values $\alpha+p$, α a given integer or fractional value, and p any integer number from $-\infty$ to $+\infty$, zero included.

Riemann deduces a theory of fractional differentiation: but without considering the question which has always appeared to me to be the great difficulty in such a theory: what is the real meaning of a complementary function containing an infinity of arbitrary constants? or, in other words, what is the arbitrariness of the complementary function of this nature which presents itself in the theory?

I wish to point out the relation between the paper referred to, and a short paper of my own "On a doubly infinite Series," *Quart. Math. Journ.* t. VI. (1851), pp. 45–47, [102]: this commences with the remark "The following completely paradoxical investigation of the properties of the function Γ (which I have been in possession

* *Werke*, pp. 331–344.

of for some years) may perhaps be found interesting from its connexion with the theories of expansion and divergent series." And I then give the expansion

$$C_n\, e^x = \sum\nolimits' [n-r]^n\, x^{n-r},$$

where n is any integer or fractional number whatever, and the summation extends to all positive and negative integer values (zero included) of r. And I remark that, n being an integer, we have $C_n = \Gamma(n)$, and hence that assuming that this is so in general, or writing

$$\Gamma(n)\, e^x = \sum\nolimits' [n-1]^r\, x^{n-r-1},$$

we have this equation as a definition of $\Gamma(n)$. The point of resemblance of course is that we have a doubly infinite expansion of e^x in a series of integer or fractional powers of x, corresponding to Riemann's like expansion of Z_{x+h} in powers of h.

Cambridge, 10 Sept. 1879.

752.

ON THE FINITE GROUPS OF LINEAR TRANSFORMATIONS OF A VARIABLE; WITH A CORRECTION.

[From the *Mathematische Annalen*, t. XVI. (1880), pp. 260–263; 439, 440.]

In the paper "Ueber endliche Gruppen linearer Transformationen einer Veränderlichen," *Math. Ann.* t. XII. (1877), pp. 23–46, Prof. Gordan gave in a very elegant form the groups of 12, 24 and 60 homographic transformations $\dfrac{ax+b}{cx+d}$. The groups of 12 and 24 are in the like form, the group of 24 thus containing as part of itself the group of 12; but the group of 60 is in a different form, not containing as part of itself the group of 12. It is, I think, desirable to present the group of 60 in the form in which it contains as part of itself Gordan's group of 12: and moreover to identify the group of 60 with the group of the 60 positive permutations of 5 letters or (writing *abc* for the cyclical permutation a into b, b into c, c into a, and so in other cases) say with the group of the 60 positive permutations 1, *abc*, *ab.cd* and *abcde*.

Any two forms of a group are, it is well known, connected as follows, viz. if 1, α, β, ... are the functional symbols of the one form, then those of the other form are 1, $\phi\alpha\phi^{-1}$, $\phi\beta\phi^{-1}$, ... (where in the case in question ϕ is a functional symbol of the like homographic form, $\phi x = \dfrac{Ax+B}{Cx+D}$). But instead of obtaining the new form in this manner, I found it easier to use the values of the rotation-symbol

$$\cos\tfrac{\theta}{2} + \sin\tfrac{\theta}{2}(i\cos X + j\cos Y + k\cos Z)$$

for the axes of the icosahedron or dodecahedron, given in my paper "Notes on polyhedra," *Quart. Math. Jour.* t. VII. (1866), pp. 304–316, [375]; viz. if for any axes, λ, μ, ν denote the parameters of rotation $\tan\frac{\theta}{2}\cos X$, $\tan\frac{\theta}{2}\cos Y$, $\tan\frac{\theta}{2}\cos Z$, then,

by a formula which is in fact equivalent to that given in my note "On the correspondence of Homographies and Rotations," *Math. Annalen*, t. XV. (1879), pp. 238–240, [660], the corresponding homographic function of x is

$$\frac{(-\nu-i)x+\lambda+i\mu}{(\lambda-i\mu)x+\nu-i},$$

where i denotes $\sqrt{-1}$ as usual.

The new formulae for the group of 60, or icosahedron group, of homographic functions $\dfrac{\alpha x+\beta}{\gamma x+\delta}$ are contained in the following table, where the four columns show the values of the coefficients α, β, γ, δ respectively: and where in the outside column, the substitution is represented as a permutation-symbol on the five letters *abcde*: moreover for shortness θ is written to denote $\sqrt{5}$.

THE GROUP OF 60.

#	α	β	γ	δ	subst.
1	1	0	0	1	1
2	-1	0	0	1	*ab.cd*
3	0	1	1	0	*ac.bd*
4	0	-1	1	0	*ad.bc*
5	2	$-3+\theta+i(1-\theta)$	$-3+\theta+i(-1+\theta)$	-2	*bc.de*
6	2	$-3+\theta+i(-1+\theta)$	$-3+\theta+i(1-\theta)$	-2	*ae.bc*
7	2	$3-\theta+i(-1+\theta)$	$3-\theta+i(1-\theta)$	-2	*ad.ce*
8	2	$3-\theta+i(1-\theta)$	$3-\theta+i(-1+\theta)$	-2	*ad.be*
9	2	$-1-\theta+i(1-\theta)$	$-1-\theta+i(-1+\theta)$	-2	*ae.cd*
10	2	$-1-\theta+i(-1+\theta)$	$-1-\theta+i(1-\theta)$	-2	*ab.de*
11	2	$1+\theta+i(-1+\theta)$	$1+\theta+i(1-\theta)$	-2	*bc.cd*
12	2	$1+\theta+i(1-\theta)$	$1+\theta+i(-1+\theta)$	-2	*ab.ce*
13	2	$-1-\theta+i(-3-\theta)$	$-1-\theta+i(3+\theta)$	-2	*ac.be*
14	2	$-1-\theta+i(3+\theta)$	$-1-\theta+i(-3-\theta)$	-2	*bd.ce*
15	2	$1+\theta+i(3+\theta)$	$1+\theta+i(-3-\theta)$	-2	*ae.bd*
16	2	$1+\theta+i(-3-\theta)$	$1+\theta+i(3+\theta)$	-2	*ae.de*
17	$-i$	i	1	1	*abc*
18	$-i$	$-i$	1	1	*aeb*
19	i	$-i$	1	1	*ade*
20	i	i	1	-1	*acd*
21	$-i$	i	1	-1	*adb*

#	α	β	γ	δ	subst.
22	i	i	1	$-i$	*abd*
23	$-i$	$-i$	1	$-i$	*bcd*
24	i	$-i$	1	i	*bdc*
25	$-1-\theta+i(3+\theta)$	2	-2	$-1-\theta+i(-3-\theta)$	*aec*
26	$1+\theta+i(3+\theta)$	2	-2	$1+\theta+i(-3-\theta)$	*ace*
27	$1+\theta+i(-3-\theta)$	2	-2	$1+\theta+i(3+\theta)$	*bcd*
28	$-1-\theta+i(-3-\theta)$	2	-2	$-1-\theta+i(3+\theta)$	*bde*
29	$-3+\theta+i(1-\theta)$	2	2	$3-\theta+i(1-\theta)$	*bce*
30	$-3+\theta+i(-1+\theta)$	2	2	$3-\theta+i(-1+\theta)$	*bec*
31	$3-\theta+i(-1+\theta)$	2	2	$-3+\theta+i(-1+\theta)$	*aed*
32	$3-\theta+i(1-\theta)$	2	2	$-3+\theta+i(1-\theta)$	*ade*
33	2	$-1-\theta+i(-1+\theta)$	$1+\theta+i(-1+\theta)$	2	*cde*
34	2	$1+\theta+i(1-\theta)$	$-1-\theta+i(1-\theta)$	2	*ced*
35	2	$-1-\theta+i(1-\theta)$	$1+\theta+i(1-\theta)$	2	*aeb*
36	2	$1+\theta+i(-1+\theta)$	$-1-\theta+i(-1+\theta)$	2	*abe*
37	$-1-\theta+i(-3-\theta)$	2	2	$1+\theta+i(-3-\theta)$	*abcde*
38	$-1-\theta+i(1-\theta)$	2	2	$1+\theta+i(1-\theta)$	*acebd*
39	$-1-\theta+i(-1+\theta)$	2	2	$1+\theta+i(-1+\theta)$	*adbec*
40	$-1-\theta+i(3+\theta)$	2	2	$1+\theta+i(3+\theta)$	*aedcb*
41	$1+\theta+i(3+\theta)$	2	2	$-1-\theta+i(3+\theta)$	*adceb*
42	$1+\theta+i(-1+\theta)$	2	2	$-1-\theta+i(-1+\theta)$	*acbde*
43	$1+\theta+i(1-\theta)$	2	2	$-1-\theta+i(1-\theta)$	*aedbc*
44	$1+\theta+i(-3-\theta)$	2	2	$-1-\theta+i(-3-\theta)$	*abecd*
45	$-1-\theta+i(-1+\theta)$	2	-2	$-1-\theta+i(1-\theta)$	*acbed*
46	$-3+\theta+i(-1+\theta)$	2	-2	$-3+\theta+i(1-\theta)$	*abdce*
47	$3-\theta+i(-1+\theta)$	2	-2	$3-\theta+i(1-\theta)$	*aecdb*
48	$1+\theta+i(-1+\theta)$	2	-2	$1+\theta+i(1-\theta)$	*adebc*
49	$1+\theta+i(1-\theta)$	2	-2	$1+\theta+i(-1+\theta)$	*aecbd*
50	$3-\theta+i(1-\theta)$	2	-2	$3-\theta+i(-1+\theta)$	*acdeb*
51	$-3+\theta+i(1-\theta)$	2	-2	$-3+\theta+i(-1+\theta)$	*abedc*
52	$-1-\theta+i(1-\theta)$	2	-2	$-1-\theta+i(-1+\theta)$	*adbce*
53	2	$-3+\theta+i(-1+\theta)$	$3-\theta+i(-1+\theta)$	2	*aebde*
54	2	$-1-\theta+i(3+\theta)$	$1+\theta+i(3+\theta)$	2	*abced*
55	2	$1+\theta+i(-3-\theta)$	$-1-\theta+i(-3-\theta)$	2	*adecb*
56	2	$3-\theta+i(1-\theta)$	$-3+\theta+i(1-\theta)$	2	*acdbe*
57	2	$-3+\theta+i(1-\theta)$	$3-\theta+i(1-\theta)$	2	*abdec*
58	2	$-1-\theta+i(-3-\theta)$	$1+\theta+i(-3-\theta)$	2	*adcbe*
59	2	$1+\theta+i(3+\theta)$	$-1-\theta+i(3+\theta)$	2	*aebcd*
60	2	$3-\theta+i(-1+\theta)$	$-3+\theta+i(-1+\theta)$	2	*acedb*

This contains (as one of five groups of 12) the group of the positive permutations of *abcd*; and, completing this into a group of 24, we have

Groups of 12 and 24.

#	α	β	γ	δ	subst.
1	1	0	0	1	1
2	-1	0	0	1	*ab.cd*
3	0	1	1	0	*ac.bd*
4	0	-1	1	0	*ad.bc*
5	$-i$	i	1	1	*abc*
6	-1	$-i$	1	i	*acb*
7	1	$-i$	1	1	*adc*
8	$-i$	$-i$	1	-1	*acd*
9	i	$-i$	1	-1	*aeb*
10	1	i	1	$-i$	*abd*
11	-1	$-i$	1	$-i$	*bcd*
12	i	$-i$	1	i	*bdc*
13	i	i	1	1	*adbc*
14	$-i$	i	1	1	*acbd*
15	1	1	1	-1	*cd*

#	α	β	γ	δ	subst.
16	-1	i	-1	1	*ab*
17	1	$-i$	1	1	*aedb*
18	$-i$	-1	$-i$	i	*bd*
19	i	1	1	i	*abcd*
20	1	-1	1	-1	*bc*
21	-1	-1	1	-1	*abdc*
22	i	-1	1	$-i$	*ac*
23	-1	i	1	$-i$	*adcb*
24	-1	1	1	1	*ad*

The groups of 60 and 24 thus each of them contain the group of 12,

$$x,\quad -\tfrac{1}{x},\quad \tfrac{1-x}{1+x},\quad \tfrac{1+x}{1-x},\quad \tfrac{x+i}{x-i},\quad \tfrac{x-i}{-x+i}.$$

It may be remarked that, to verify the periodicities of the forms contained in the group of 60, we have as the conditions that

$$\frac{\alpha x+\beta}{\gamma x+\delta} \text{ may be periodic of the order 2, } \frac{\alpha+\delta}{\alpha\delta-\beta\gamma}=0, \text{ that is, } \alpha+\delta=0,$$

$$\begin{array}{llllll} ,, & ,, & ,, & 3, & ,, & =1, \\ ,, & ,, & ,, & 5, & ,, & =\tfrac{1}{2}(3+\sqrt{5}). \end{array}$$

For instance, in the form

$$\frac{[-1-\Theta+i(-3-\Theta)]\,x+2}{2x+[1+\Theta+i(-3-\Theta)]},$$

we have

$$\alpha\delta=-(1+\Theta)^2-(3+\Theta)^2\,i^2=-2\Theta-8\Theta,\quad \beta\gamma=4,$$
$$\alpha+\delta=-2i(3+\Theta);$$

and therefore

$$\frac{(\alpha+\delta)^2}{\alpha\delta-\beta\gamma}=\frac{-4(3+\Theta)^2}{-3(3+\Theta)},\quad =\frac{3+\Theta}{2}=\tfrac{1}{2}(3+\sqrt{5}),$$

as it should be.

Cambridge, 11 Nov. 1879.

Correction*, pp. 439, 440.

I erroneously assumed that the symbol *adcb* could be taken as corresponding to the linear transformation ix: but this was obviously wrong, for it gave *bd* as corresponding to the transformation $-ix$, and these are not of the same order, but of the orders 4 and 2 respectively. The proper symbol is *adbc*, as given above, and the remaining eleven symbols are then at once obtained.

Cambridge, 17 Feb. 1880.

* The correction in the Table of the Groups of 12 and 24 has been inserted in the Table as now printed on p. 240; it applies to the second half of the column of symbols on the extreme right-hand.

753.

ON A THEOREM RELATING TO THE MULTIPLE THETA-FUNCTIONS.

[From the *Mathematische Annalen*, t. XVII. (1880), pp. 115–122.]

I PROPOSE—partly for the sake of the theorem itself, partly for that of the notation which will be employed—to demonstrate the general theorem (3′), p. 4, of Dr Schottky's *Abriss einer Theorie der Abel'schen Functionen von drei Variabeln*, (Leipzig, 1880), which theorem is there presented in the form:

$$e^{-H(u_1,\ldots;\mu',\nu')}\,\Theta(u_1+2\varpi_1',\ldots;\mu,\nu)=e^{-2\pi i\mu\nu'}\,\Theta(u_1,\ldots;\mu+\mu',\nu+\nu'), \tag{3'}$$

but which I write in the slightly different form

$$\exp.[-H(u;\mu',\nu')]\cdot\Theta(u+2\varpi';\mu,\nu)=\exp.[-2\pi i\mu\nu']\cdot\Theta(u;\mu+\mu',\nu+\nu').$$

I remark that the theorem is given in the preliminary paragraphs the contents of which are, as mentioned by the Author, derived from Herr Weierstrass: and that the form of the theta-function is a very general one, depending on the general quadric function

$$G(u_1,\ldots,u_p;u_1,\ldots,u_p)$$

of $2p$ variables, p being the number of the arguments $u_1,\ldots,u_p$ (in fact, the periods are not reduced to the normal form, but are arbitrary); and the characters $\nu_1,\ldots,\nu_p$, $\mu_1,\ldots,\mu_p$, instead of having each of them the value 0 or 1, have each of them any integer or fractional value whatever. The meaning of the theorem (u denoting a set or row of p letters $u_1,\ldots,u_p$, and so in other cases), is that the function

$$\Theta(u;\mu+\mu',\nu+\nu')$$

with the new characters $\mu+\mu'$ and $\nu+\nu'$ is, save as to an exponential factor, equal to the function $\Theta(u+2\varpi';\mu,\nu)$ with the original characters μ, ν, but with the new arguments $u+2\varpi'$.

Notation.

This is in some measure a development of the notation employed in my "Memoir on the Theory of Matrices," *Phil. Trans.* t. CXLVIII. (1858), pp. 17–37, [152]: I use certain single letters u, etc. to denote sets or rows each of p letters, $u=(u_1,\ldots,u_p)$: or if, to fix the ideas $p=3$, then $u=(u_1,u_2,u_3)$, and so in other cases.

But I use certain other letters a, etc. to denote squares or matrices each of p^2 letters; thus, if $p=3$ as before,

$$a=\begin{vmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{vmatrix},$$

and in any such case the transposed matrix is denoted by the same letter enclosed in parentheses

$$(a)=\begin{vmatrix}a_{11}&a_{21}&a_{31}\\a_{12}&a_{22}&a_{32}\\a_{13}&a_{23}&a_{33}\end{vmatrix}.$$

The sum $u+v$ of the row-letters $u, = (u_1, u_2, u_3)$ and $v, = (v_1, v_2, v_3)$ denotes the row $(u_1+v_1, u_2+v_2, u_3+v_3)$: and in like manner the sum $a+b$ of the two matrices, or square-letters a and b, denotes the matrix

$$\begin{vmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \\ a_{31}+b_{31} & a_{32}+b_{32} & a_{33}+b_{33} \end{vmatrix},$$

and similarly for a sum of three or more terms.

The product $uv, = (u_1, u_2, u_3)(v_1, v_2, v_3)$, of the two row-letters u, v denotes the single term $u_1v_1 + u_2v_2 + u_3v_3$. We have $uv = vu$.

The product

$$au, = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} (u_1, u_2, u_3),$$

of a preceding square-letter a and a succeeding row-letter u, denotes the set or row

$$(a_{11}, a_{12}, a_{13})(u_1, u_2, u_3),\ (a_{21}, a_{22}, a_{23})(u_1, u_2, u_3),\ (a_{31}, a_{32}, a_{33})(u_1, u_2, u_3);$$

the notation ua is not employed.

The product

$$auv = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} (u_1, u_2, u_3)(v_1, v_2, v_3),$$

of a preceding square-letter a followed by the two row-letters u and v, denotes the single term

$$\begin{aligned} &(a_{11}, a_{12}, a_{13})(u_1, u_2, u_3)\, v_1 \\ +\, &(a_{21}, a_{22}, a_{23})(u_1, u_2, u_3)\, v_2 \\ +\, &(a_{31}, a_{32}, a_{33})(u_1, u_2, u_3)\, v_3. \end{aligned}$$

Observe that auv is not in general $= avu$; but it is easy to verify that $auv = (a)vu$; and hence if $(a) = a$, that is, if the matrix a be symmetrical, then $auv = avu$.

A product of two matrices

$$ab, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix},$$

denotes a matrix

$$\begin{vmatrix} (b_{11}, b_{21}, b_{31}) & (b_{12}, b_{22}, b_{32}) & (b_{13}, b_{23}, b_{33}) \\ (a_{11}, a_{12}, a_{13}) & \cdots & \cdots \\ (a_{21}, a_{22}, a_{23}) & \cdots & \cdots \\ (a_{31}, a_{32}, a_{33}) & \cdots & \cdots \end{vmatrix};$$

viz. the top-line of the compound matrix is

$$(a_{11}, a_{12}, a_{13})(b_{11}, b_{21}, b_{31}),\ (a_{11}, a_{12}, a_{13})(b_{12}, b_{22}, b_{32}),\ (a_{11}, a_{12}, a_{13})(b_{13}, b_{23}, b_{33}),$$

and so for the other lines: or expressing this in words, we say that any line of the compound matrix is obtained by compounding the corresponding line of the first or further component matrix with the several columns of the second or nearer component matrix.

Clearly ab is not in general $= ba$. We may easily verify that $(ab) = (b)(a)$, that is, the transposed matrix (ab) is that obtained by the composition of the transposed matrix (b) as first or further matrix, with the transposed matrix (a) as second or nearer matrix. Even if a and b

are each symmetrical, we do not in general have $ab = ba$, but only $(ab) = ba$, or what is the same thing, $ab = (ba)$.

In a symbol such as $abuv$, we first combine a, b into a single matrix ab, and then regard the expression as a combination such as auv: the expression denotes therefore a single term. The theory might be explained in greater detail; but the mode of working with row- and square-letters will be readily understood from what precedes.

In all that follows, u, μ, ν, μ', ν', ϖ, ϖ', ζ' are row-letters; a, b, h, ω, ω', η, η' are square-letters: a and b are symmetrical, viz. $a = (a)$, $b = (b)$.

And I write

$$\begin{aligned}(\cdot)(u,v)^2 &= (a,h,b)(u,v)^2 \\ &= au^2 + 2huv + bv^2\end{aligned}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} (u_1, u_2, u_3)^2$$

$$+2 \begin{vmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{vmatrix} (u_1, u_2, u_3)(v_1, v_2, v_3)$$

$$+ \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix} (v_1, v_2, v_3)^2$$

to denote the general quadric function of the $2p$ letters u, v, with

$$\tfrac{1}{2}p(p+1) + p^2 + \tfrac{1}{2}p(p+1), \quad = p(2p+1)$$

coefficients. It is assumed that the determinant formed with the $\frac{1}{2}p(p+1)$ coefficients b is negative: this is the necessary and sufficient condition for the convergence of the series.

Definition of $\Theta(u;\mu,\nu)$.

$\Theta(u;\mu,\nu)$, the general theta-function with p arguments u, and $2p$ characters μ, ν, is the sum of a p-tuple series of exponentials

$$\Theta(u;\mu,\nu) = \sum \exp.[(\cdot)(n, n+\nu)^2 + 2\pi i u\,(n+\nu)],$$

where each of the letters n, $= (n_1, \ldots, n_p)$, has all integer values (zero included) from $-\infty$ to $+\infty$.

The general theorem in regard to $\Theta(u;\mu,\nu)$.

This is

$$\exp.[-H(u;\mu',\nu')] \cdot \Theta(u + 2\varpi';\mu,\nu) = \exp.[-2\pi i \mu\nu'] \cdot \Theta(u;\mu+\mu',\nu+\nu'),$$

establishing a relation between the function $\Theta(u;\mu+\mu',\nu+\nu')$, with arbitrary character-increments μ', ν', and the function $\Theta(u+2\varpi';\mu,\nu)$ with the original characters, but with new arguments $u + 2\varpi'$. Also $H(u;\mu',\nu')$ denotes a function, linear as regards the arguments u, but quadric as regards μ' and ν'; $-2\pi i \mu\nu'$ is a single term depending only on μ and ν'; and the theorem thus is that the two functions differ only by an exponential factor. The relations between the constants will be obtained in the course of the investigation.

Demonstration.

The truth of the theorem depends on the equality of corresponding exponentials on the two sides of the equation: viz. substituting for the theta-functions their values, and comparing the exponents or arguments of the exponentials: writing also for convenience

$$G(u+2\varpi', n+\nu),$$

to denote the quadric function $(\cdot)(u+2\varpi', n+\nu)^2$; we ought to have

$$\begin{aligned} &-H(u;\mu',\nu') + G(u+2\varpi', n+\nu) + 2\pi i u(n+\nu) \\ &\quad = -2\pi i\mu\nu' + G(u, n+\nu+\nu') + 2\pi i(\mu+\mu')(n+\nu+\nu'), \end{aligned}$$

or say

$$H(u;\mu',\nu') = G(u+2\varpi', n+\nu) - G(u, n+\nu+\nu') - 2\pi i(n+\nu+\nu')\mu'.$$

In this equation, if true at all, the terms containing n must destroy each other; assuming that they do so, the equation becomes

$$H(u;\mu',\nu') = G(u+2\varpi', \nu) - G(u, \nu+\nu') - 2\pi i(\nu+\nu')\mu'.$$

Consider first the terms in n: the right-hand side is

$$\begin{aligned} &= a(u+2\varpi')^2 + 2h(u+2\varpi')(n+\nu) + b(n+\nu)^2 \\ &\quad - au^2 - 2hu(n+\nu+\nu') - b(n+\nu+\nu')^2 - 2\pi i n\mu', \end{aligned}$$

and the terms herein which contain n thus are

$$\begin{aligned} &2h(u+2\varpi')n + bn^2 + 2bn\nu \\ &\quad - 2hun - bn^2 - 2bn(\nu+\nu') - 2\pi i n\mu', \\ &= 4h\varpi' n - 2bn\nu' - 2\pi i n\mu', \end{aligned}$$

which, b being symmetrical, may be written

$$= 2(2h\varpi' - b\nu' - \pi i\mu')\, n,$$

and these terms will vanish if, and only if

$$2h\varpi' - b\nu' - \pi i\mu' = 0,$$

a system of p equations connecting ϖ', μ', ν'.

Assuming them to be satisfied, the remaining relation,

$$H(u;\mu',\nu') = G(u+2\varpi', \nu) - G(u, \nu+\nu') - 2\pi i(\nu+\nu')\mu',$$

becomes

$$\begin{aligned} H(u;\mu',\nu') &= a(u+2\varpi')^2 + 2h(u+2\varpi')\nu + b\nu^2 \\ &\quad - au^2 - 2hu(\nu+\nu') - b(\nu+\nu')^2 - 2\pi i(\nu+\nu')\mu'. \end{aligned}$$

Here, a and b being symmetrical, we have

$$a(u+2\varpi')^2 = au^2 + 4a\varpi' u + 4a\varpi'^2, \quad b(\nu+\nu')^2 = b\nu^2 + 2b\nu'\nu + b\nu'^2,$$

and the value therefore is

$$= 4a(\varpi' u + \varpi'^2) + 2h(2\varpi'\nu - u\nu') - b(2\nu'\nu + \nu'^2) - 2\pi i(\nu+\nu')\mu'.$$

On the right-hand side, putting the term in h under the form

$$-2h(u+\varpi')\nu' + 2h\varpi'(2\nu+\nu'), \quad = -2(h)\nu'(u+\varpi') + 2h\varpi'(2\nu+\nu'),$$

and the last term under the form

$$-\pi i\mu'(2\nu+\nu') - \pi i\mu'\nu',$$

the equation becomes

$$\begin{aligned} H(u;\mu',\nu') = {} & (4a\varpi' - 2(h)\nu')(u+\varpi') - \pi i\mu'\nu' \\ & + (2h\varpi' - b\nu' - \pi i\mu')(2\nu+\nu'), \end{aligned}$$

where the second line vanishes in virtue of the foregoing equation

$$2h\varpi' - b\nu' - \pi i\mu' = 0;$$

the equation thus is

$$H(u;\mu',\nu') = (4a\varpi' - 2(h)\nu')(u+\varpi') - \pi i\mu'\nu',$$

which equation, regarding therein ϖ' as a linear function of μ' and ν', shows that $H(u;\mu',\nu')$ is a function linear as regards u (and containing this only through $u+\varpi'$), but quadric as regards μ', ν'.

Introducing the new row-letter ζ', we may write

$$H(u;\mu',\nu') = 2\zeta'(u+\varpi') - \pi i\mu'\nu',$$

viz. the expression on the right-hand side is here assumed as the value of the function

$$H(u;\mu',\nu'), \quad = G(u+2\varpi',\nu) - G(u,\nu+\nu') - 2\pi i(\nu+\nu')\mu',$$

and the theorem then is

$$\exp.[-H(u;\mu',\nu')]\cdot\Theta(u+2\varpi';\mu,\nu) = \exp.[-2\pi i\mu\nu']\cdot\Theta(u;\mu+\mu',\nu+\nu'),$$

where, by what precedes,

$$\begin{aligned} 2h\varpi' - b\nu' - \pi i\mu' &= 0, \\ 2a\varpi' - (h)\nu' - \zeta' &= 0, \end{aligned}$$

$2p$ equations for determining the $2p$ functions ϖ', ζ' as linear functions of μ', ν': which equations depend on the $p(2p+1)$ constants a, b, h.

Suppose that the resulting values of ϖ' and ζ' are

$$\varpi' = \omega\mu' + \omega'\nu',$$

$$\zeta' = \eta\mu' + \eta'\nu',$$

where ω, ω', η, η' are square-letters; then, regarding a, b, h as arbitrary, the $4p^2$ new constants ω, ω', η, η' cannot be all of them arbitrary, but must be connected by $4p^2-p(2p+1)$, $=p(2p-1)$ equations.

We may regard ω, ω', η, η' as satisfying these $p(2p-1)$ equations, but as being otherwise arbitrary; the foregoing equations then are

$$\begin{aligned} 2h\varpi' - b\nu' - \pi i\mu' &= 0, \\ 2a\varpi' - (h)\nu' - \zeta' &= 0, \\ \varpi' &= \omega\mu' + \omega'\nu', \\ \zeta' &= \eta\mu' + \eta'\nu', \end{aligned}$$

which lead to the equations connecting a, b, h with ω, ω', η, η'.

The first and second equations, substituting for ϖ' and ζ' their values, become

$$(2h\omega - \pi i)\mu' + (2h\omega' - b)\nu' = 0,$$
$$(2a\omega - \eta)\mu' + (2a\omega' - (h) - \eta')\nu' = 0,$$

or μ', ν' being arbitrary, we thus obtain the $4p^2$ equations

$$\begin{aligned} 2a\omega - \eta &= 0, \\ 2h\omega - \pi i &= 0, \\ 2a\omega' - \eta' - (h) &= 0, \\ 2h\omega' - b &= 0, \end{aligned}$$

which are the equations in question. It is to be observed that in πi is, like the other symbols, a matrix, viz. it is regarded as containing the matrix unity; or, what is the same thing, it denotes

$$\pi i \begin{vmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ \vdots & & \ddots & \end{vmatrix}.$$

We can eliminate a, b, h from these equations and thus obtain the $p(2p-1)$ equations before referred to, which connect the $4p^2$ constants ω, ω', η, η'. I give, but without a complete explanation, the steps of the elimination.

The equation $2a\omega - \eta = 0$, may be written in the form

$$2(a\omega) - (\eta) = 0,$$

that is,

$$2(\omega)(a) - (\eta) = 0,$$

or since $(a) = a$, this is

$$2(\omega)a - (\eta) = 0;$$

from the original form, and the new form respectively, we find

$$2(\omega)a\omega - (\omega)\eta = 0, \quad 2(\omega)a(\omega) - (\eta)\omega = 0;$$

and comparing these

$$(\omega)\eta - (\eta)\omega = 0, \quad \text{(first result)}.$$

The equation $2a\omega' - \eta' - (h) = 0$, or say $(h) = -\eta' + 2a\omega'$, may be written in the form

$$h = -(\eta') + 2(a\omega'),$$

that is, since $a = (a)$,

$$h = -(\eta') + 2(\omega')a;$$

and we thence deduce

$$h\omega = -(\eta')\omega + 2(\omega')a\omega.$$

But from the equation $2a\omega - \eta = 0$, we have $2(\omega')a\omega - (\omega')\eta = 0$, and the equation thus becomes $h\omega = -(\eta')\omega + (\omega')\eta$; which, in virtue of $2h\omega - \pi i = 0$, becomes

$$\tfrac{1}{2}\pi i = -(\eta')\omega + (\omega')\eta, \quad \text{(second result)}.$$

From the equation above obtained, $h = -(\eta') + 2(\omega')a$, we have

$$h\omega' = -(\eta')\omega' + 2(\omega')a\omega';$$

in virtue of $2h\omega' - b = 0$, this becomes $-2(\eta')\omega' + 4(\omega')a\omega' = b$; an equation which may also be written $-2((\eta')\omega') + 4((\omega')a\omega') = (b)$, or, what is the same thing, $-2(\omega')\eta' + 4(\omega')(a)\omega' = (b)$; or since $(a) = a$ and $(b) = b$, this is

$$-2(\omega')\eta' + 4(\omega')a\omega' = b:$$

and comparing with the original equation

$$-2(\eta')\omega' + 4(\omega')a\omega' = b,$$

we obtain

$$(\omega')\eta' - (\eta')\omega' = 0, \quad \text{(third result)}.$$

We have thus the three systems

$$\begin{aligned} (\omega)\eta - (\eta)\omega &= 0, \quad \tfrac{1}{2}p(p-1) \text{ equations,} \\ (\omega')\eta - (\eta')\omega &= \tfrac{1}{2}\pi i, \quad p^2 \text{ equations,} \\ (\omega')\eta' - (\eta')\omega' &= 0, \quad \tfrac{1}{2}p(p-1) \text{ equations,} \end{aligned}$$

in all $p(2p-1)$ equations. As to these systems, observe that $(\omega)\eta$, $(\eta)\omega$, etc., are all of them matrices of p^2 terms; each of the three systems denotes therefore in the first instance p^2 equations, viz. the equations obtained by equating to zero the several terms of such a matrix: but in the first system each diagonal term so equated to zero gives the identity $0 = 0$; and equating to zero the terms which are symmetrical in regard to the diagonal we obtain twice over, in the forms $P = 0$, and $-P = 0$, one and the same equation; the number of equations is thus diminished from p^2 to $\frac{1}{2}p(p-1)$; and similarly in the third system the number of equations is $= \frac{1}{2}p(p-1)$: but for the second system the number of equations is really $= p^2$. It is hardly necessary to remark that in this second system $\frac{1}{2}\pi i$ is as before regarded as a matrix.

The foregoing three systems of equations are in fact the equations (6) p. 4 of Dr Schottky's work.

Cambridge, 12 July, 1880.

754.

ON THE CONNEXION OF CERTAIN FORMULÆ IN ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. IX. (1880), pp. 23–25.]

In reference to a like question in the theory of the double ϑ-functions, it is interesting to show that (if not completely, at least very nearly) the single formula

$$\Pi(u, a) = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta(u-a)}{\Theta(u+a)},$$

that is,

$$\int_0^u \frac{k^2 \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a \operatorname{sn}^2 u\, du}{1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u} = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta(u-a)}{\Theta(u+a)},$$

leads not only to the relation

$$\log\Theta u = \tfrac{1}{2}\log\frac{2k'K}{\pi} + \tfrac{1}{2}\left(1 - \frac{E}{K}\right)u^2 - k^2\int_0^u du \int_0^u du \operatorname{sn}^2 u,$$

between the functions Θ, sn, but also to the addition-equation for the function sn.

Writing in the equation a indefinitely small, and assuming only that $\operatorname{sn} a$, $\operatorname{cn} a$, $\operatorname{dn} a$ then become a, 1, 1, respectively, the equation is

$$k^2 a \int_0^u \operatorname{sn}^2 u\, du = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta u - a\Theta' u}{\Theta u + a\Theta' u} = u\frac{\Theta' 0}{\Theta 0} - a\frac{\Theta' u}{\Theta u},$$

that is,

$$\frac{\Theta' u}{\Theta u} = u\,\frac{\Theta'' 0}{\Theta 0} - k^2\int_0^u du\, \operatorname{sn}^2 u,$$

or, integrating from $u = 0$, this is

$$\log\Theta u = C + \tfrac{1}{2}u^2\,\frac{\Theta'' 0}{\Theta 0} - k^2\int_0^u du \int_0^u du\, \operatorname{sn}^2 u,$$

which, except as regards the determination of the constants, is the required equation for $\log\Theta u$.

Next, differentiating twice the equation for $\Pi(u, a)$, and once the equation obtained for $\frac{d\Theta'}{du}$, we have

$$k^2 \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a\,\frac{d}{du}\left(\frac{\operatorname{sn}^2 u}{1 - k^2\operatorname{sn}^2 a\operatorname{sn}^2 u}\right) = \tfrac{1}{2}\,\frac{\Theta''\Theta - \Theta'^2}{\Theta^2}(u-a) - \tfrac{1}{2}\,\frac{\Theta''\Theta - \Theta'^2}{\Theta^2}(u+a),$$

and

$$\frac{\Theta''\Theta - \Theta'^2}{\Theta^2} = \frac{\Theta'' 0}{\Theta 0} - k^2\operatorname{sn}^2 u,$$

where, for shortness, $\frac{\Theta'' 0}{\Theta 0}$ is written to denote $\frac{\Theta'' u\,\Theta u - (\Theta' u)^2}{\Theta u^2}$, and the like in the first equation; the right-hand side of the first equation therefore is

$$= -\tfrac{1}{2}k^2\left\{\operatorname{sn}^2(u-a) - \operatorname{sn}^2(u+a)\right\},$$

or the equation becomes

$$2\operatorname{sn} a \operatorname{cn} a \operatorname{dn} a\,\frac{d}{du}\,\frac{\operatorname{sn}^2 u}{1 - k^2\operatorname{sn}^2 u\operatorname{sn}^2 a} = \operatorname{sn}^2(u+a) - \operatorname{sn}^2(u-a),$$

that is,

$$\frac{4\operatorname{sn} u\operatorname{sn}' u\operatorname{sn} a\operatorname{cn} a\operatorname{dn} a}{(1 - k^2\operatorname{sn}^2 u\operatorname{sn}^2 a)^2} = \operatorname{sn}^2(u+a) - \operatorname{sn}^2(u-a).$$

The numerator on the left-hand side must be a symmetrical function of u, a, and hence (even if the value of $\operatorname{sn}' u$ were unknown) it would appear that $\operatorname{sn}' u$ must be a mere constant multiple of $\operatorname{cn} u \operatorname{dn} u$; assuming, however, the actual value, $\operatorname{sn}' u = \operatorname{cn} u\operatorname{dn} u$, the formula is

$$\frac{4\operatorname{sn} u\operatorname{cn} u\operatorname{dn} u\operatorname{sn} a\operatorname{cn} a\operatorname{dn} a}{(1 - k^2\operatorname{sn}^2 u\operatorname{sn}^2 a)^2} = \operatorname{sn}^2(u+a) - \operatorname{sn}^2(u-a) = \{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}\{\operatorname{sn}(u+a) - \operatorname{sn}(u-a)\}.$$

The factor $\{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}$ becomes $= 2\operatorname{sn} u$ for $a = 0$, and this suggests that the factor $\operatorname{sn} u$ on the left-hand side is a factor of $\{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}$. That $\operatorname{cn} u$ is not a factor hereof would follow from the properties of the period K; viz. for $u = K$, $\operatorname{cn} u = 0$, but $\{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}, = 2\operatorname{sn}(K+a)$ is not $= 0$; and, similarly, that $\operatorname{dn} u$ is not a factor from the properties of the period iK'; hence, $\operatorname{cn} u$, $\operatorname{dn} u$ belong to the other factor $\{\operatorname{sn}(u+a) - \operatorname{sn}(u-a)\}$, and by symmetry $\operatorname{cn} a$, $\operatorname{dn} a$ belong to the first-mentioned factor. And we are thus led to assume

$$\begin{aligned}\operatorname{sn}(u+a) + \operatorname{sn}(u-a) &= 2M\operatorname{sn} u\operatorname{cn} a\operatorname{dn} a,\\ \operatorname{sn}(u+a) - \operatorname{sn}(u-a) &= 2M'\operatorname{sn} a\operatorname{cn} u\operatorname{dn} u,\end{aligned}$$

where

$$\text{denom.} = 1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u,$$

and $MM' = 1$. Some further investigation is wanting to show that M and M' are constants, but assuming that they are so and each $= 1$, the formulae give at once the ordinary expression for $\operatorname{sn}(u + a)$; that is, we have the addition-equation for the function sn.

[Paper 754 continues to PDF p. 276 / printed p. 252.]